ENTERPRISE EDTECH ARCHITECTURE

CBSE & NEP 2020 ALIGNED

AI CLASSROOM COMPANION

SAVRA AI

The Intelligent AI Teaching Companion Built for Indian Classrooms

> Comprehensive Technical Blueprint, Full-Stack Architecture, Data Schema, Pedagogical AI Engine, User Journeys, REST API Catalog & System Specifications.

PLATFORM CORE

Next.js 16 â€¢ Express.js â€¢ Prisma ORM â€¢ PostgreSQL

AI FOUNDATION

Google Gemini 2.5 Flash â€¢ In-Context Framework Grounding

TARGET AUDIENCE

Indian K-12 Schools (Grades 1-12, Teachers, Students, Admins)

DOCUMENT VERSION & DATE

v1.0.0 Production Release â€¢ October 2026

Document Contents & Executive Overview

This document serves as the authoritative, end-to-end technical specification and architectural manual for the **Savra AI Educational Technology Platform**.

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10x Faster
LESSON PLAN CREATION

8+ Hours
SAVED WEEKLY PER TEACHER

Seconds
CBSE QP GENERATION

1. Executive Summary & Core Identity

1.1 Project Overview: What is Savra AI?

Savra AI is an enterprise-grade Educational Technology (EdTech) platform custom-engineered for Indian K-12 school ecosystems (Grades 1 through 12). Designed to address the immense administrative and cognitive burden borne by educators, Savra AI acts as an intelligent, real-time teaching companion. It empowers educators to generate curriculum-aligned lesson plans, standardized question papers, gamified classroom quizzes, and targeted announcements in seconds rather than hours.

Unlike generic AI conversational tools, Savra AI is deeply grounded in the Indian national curriculum frameworks: the **National Education Policy (NEP 2020)**, the **National Curriculum Framework (NCF 2023)**, **Bloomâ€™s Revised Taxonomy**, and the **5E Instructional Model (Engage, Explore, Explain, Elaborate, Evaluate)**. It integrates official CBSE Class 10 and Class 12 blueprint patterns directly into its LLM context window, ensuring that every piece of educational material generated conforms to national board assessment guidelines.

> **Core Platform Mission:** To bridge the gap between ambitious national educational reforms (NEP 2020/NCF 2023) and daily classroom reality by providing teachers with an intuitive, AI-powered pedagogical co-pilot that automates paperwork, personalizes learning, and unlocks deeper student engagement.

1.2 Key Platform Capabilities

âš¡ Automated Lesson Planning

Generates period-by-period lesson plans mapped across 8 distinct pedagogical dimensions: concepts, learning outcomes, 5E teaching methodology, formative assessments, teaching resources, 21st-century skills & values, real-life applications, and reflective prompts.

ðŸ“ CBSE Question Paper Generator

Produces complete examination papers across 9 distinct question types (MCQs, Case Studies, Assertion-Reasoning, Numericals, Diagrams, etc.) with automatic sectional division, general instructions, and instant concise answer keys.

ðŸŽ® Gamified Quiz Engine

Enables teachers to launch interactive classroom quizzes with timed countdowns, auto-scoring, coins rewards, and class-wide and school-wide leaderboards designed for low-device environments.

ðŸ¤– Ask Savra Multimodal AI Assistant

Role-aware generative assistant with custom personas for students (Socratic hints, curriculum guidance without cheating) and teachers (lesson design, pedagogy advice, and diagram/math image OCR analysis).

2. The Problem Landscape: What Problem Savra AI Solves

The Indian K-12 schooling system is one of the largest and most complex educational environments in the world, serving over 250 million students across public and private institutions. Despite recent transformative reforms, educators and school systems face severe systemic challenges that directly impede educational quality.

2.1 The Crisis of Teacher Administrative Burnout

In standard Indian schools, teachers spend up to **40% to 50% of their working hours on administrative documentation** rather than direct classroom instruction or student mentorship:

- **Manual Lesson Planning:** Crafting a single multi-period lesson plan compliant with institutional guidelines typically takes **40 to 60 minutes** per subject per week. A teacher managing multiple grades spends 8 to 12 hours weekly typing repetitive plan documents.
- **Repetitive Question Paper Drafting:** Sourcing questions, balancing mark distributions, ensuring question type diversity, and formatting Word documents consumes hours before every unit test, midterm, and terminal examination.
- **Rote Grading & Record Keeping:** Manually scoring objective quizzes and updating mark registers drains teacher energy that could otherwise be directed toward targeted remedial interventions.

2.2 The NEP 2020 & NCF 2023 Implementation Gap

The Government of Indiaâ€™s **National Education Policy 2020 (NEP 2020)** and the **National Curriculum Framework for School Education (NCF 2023)** mandate a dramatic shift from rote memorization toward *Competency-Based Learning (CBL)*, Higher-Order Thinking Skills (HOTS), and experiential pedagogy. However, a massive implementation gap persists:

NEP / NCF Mandate	Traditional Classroom Reality	Savra AI Algorithmic Solution
Competency-Based Assessments	Teachers rely on legacy guidebooks with outdated factual recall questions.	Generates application-heavy, case-study, and assertion-reasoning questions directly mapped to learning competencies.
5E Instructional Pedagogy	Lesson plans are written as flat summaries of textbook paragraphs without pedagogical staging.	Synthesizes Engage, Explore, Explain, Elaborate, and Evaluate steps into every single classroom period automatically.
Bloom's Taxonomy Cognitive Hierarchy	Over 80% of teacher-made tests assess solely Level 1 (Remembering) and Level 2 (Understanding).	Explicitly balances questions across Analyzing, Evaluating, and Creating cognitive tiers.
21st-Century Skills & Real-Life Context	Curriculum feels detached from daily life; teachers lack time to research modern contextual applications.	Each period plan embeds value education, critical thinking skills, and tangible Indian real-world applications.

2.3 Student Disengagement & Feedback Latency

Traditional homework and paper worksheets suffer from high submission friction and protracted feedback loops. Students receive graded papers days or weeks after submission, by which time the learning moment has passed. Savra AI solves this by introducing **instant gamified evaluation**: students attempt interactive quizzes on web or mobile, receive immediate score breakdowns with answer justifications, and earn coins that drive competitive leaderboard standings.

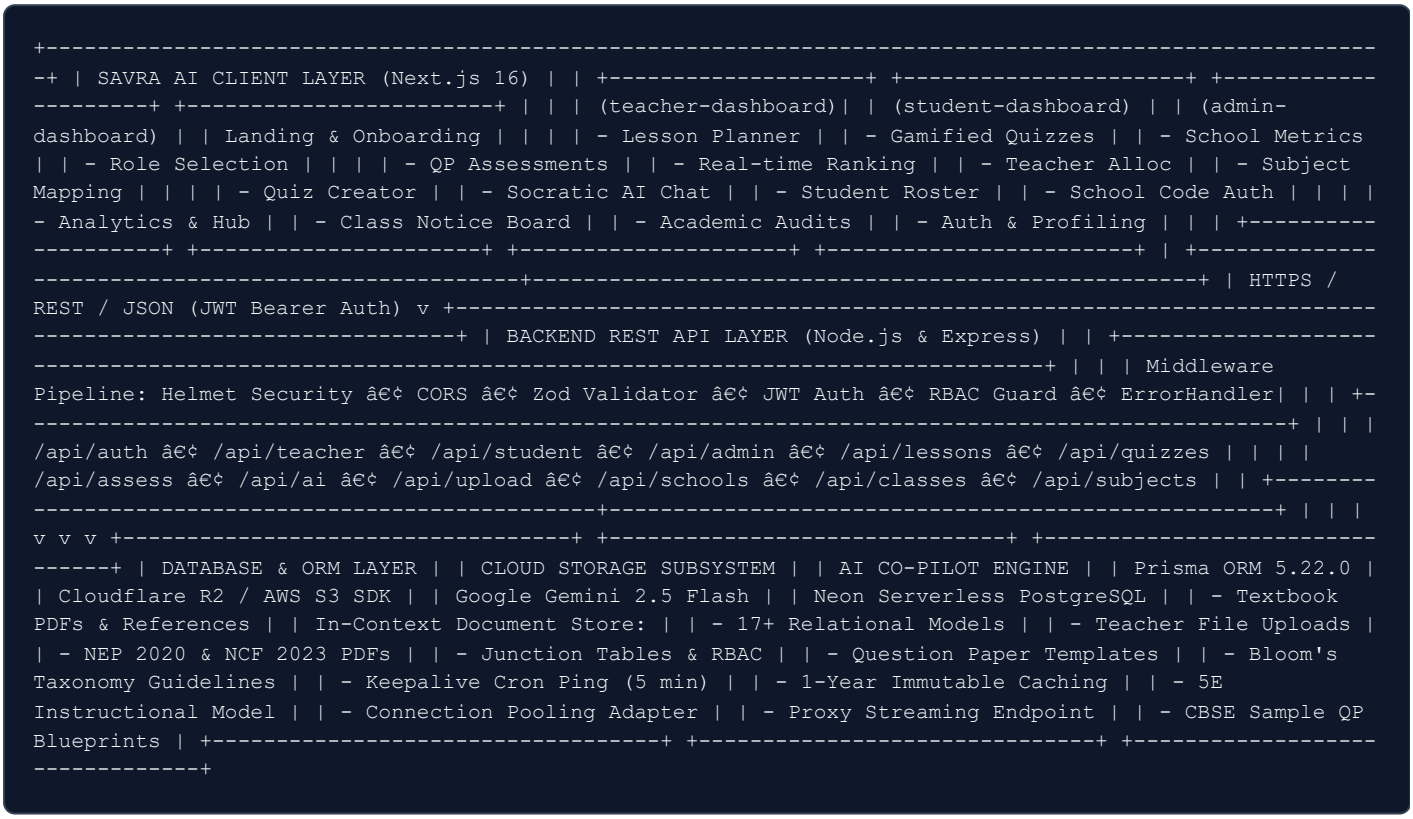
2.4 Fragmented EdTech Silos vs. Unified Platform

Prior to Savra AI, schools resorted to a disjointed patchwork of point solutions: one tool for quiz hosting (e.g., Kahoot), another for word processing lesson plans, a third for notice boards (WhatsApp groups), and separate generic AI bots (ChatGPT) that lack curriculum alignment and produce foreign curriculum outputs. Savra AI consolidates the entire teaching and learning lifecycle into a **single unified, multi-tenant portal** tailored to the Indian educational landscape.

3. High-Level System Architecture & Flow

3.1 Architectural Paradigm & Philosophy

Savra AI is architected as a high-performance, decoupled client-server platform adhering to **Domain-Driven Design (DDD)** and **Layered Service Architecture**. The platform is designed for sub-second database transactions, horizontal scalability, serverless operational efficiency, and deterministic AI generation pipelines.



3.2 Multi-Tenant Institutional Hierarchy

The database and permission model enforce a clean, multi-tenant hierarchy guaranteeing strict data isolation between schools while enabling cross-classroom analytics:

ðŸ« School Domain

The top-level institutional entity identified by a unique school code (e.g., `SAVRA001`). Houses all administrators, classrooms, teachers, and enrolled students.

ðŸ“š Academic Structure

Classes span Grades 1 through 12, each segmented into sections (A through E). Subjects are linked to standardized national curricula and granular chapter hierarchies.

ðŸ‘¥ Teacher Allocation

Teachers are mapped via many-to-many relations (`TeacherSubject` and `TeacherClass`), allowing cross-grade subject teaching and multiple classroom supervision.

ðŸŽ“ Student Enrollment

Students belong to a specific classroom roster, linked by roll numbers, tracking cumulative gamification points (`totalPoints`) and historical quiz submissions.

4. Full-Stack Technology Breakdown

Savra AI is constructed using a modern, resilient, type-safe stack designed for zero runtime exceptions, exceptional rendering performance, and secure data handling.

4.1 Frontend Architecture & Dependencies

Technology / Library	Version	Functional Role & Architectural Responsibility
Next.js (App Router)	`16.1.1`	Core React framework powering server-side routing, route grouping `(teacher-dashboard)`, `(student-dashboard)`, `(admin-dashboard)`, font optimization, and SEO metadata.
React & React DOM	`19.2.3`	Modern component lifecycle, concurrent rendering, server actions, and declarative UI composition.
TypeScript	`^5.0.0`	End-to-end type safety across client interfaces, API response contracts, and state payloads.
Tailwind CSS	`^4.0.0`	Next-gen utility-first styling with PostCSS, dynamic theme token variables, responsive design, and CSS animations.
Radix UI Primitives	`Latest`	Headless, accessible UI primitives: Dialogs, Alert Dialogs, Dropdown Menus, Select, Tabs, and Slot delegation.
TipTap Rich Text Editor	`^3.16.0`	Interactive in-browser lesson plan editing with StarterKit and placeholder extensions for seamless document customization.
Recharts	`^3.6.0`	Interactive SVG-based data visualization for student analytics, quiz completion curves, and score distributions.
Document Export (jspdf, docx)	`jspdf ^4.0, docx ^9.5`	Client-side compilation and download of formatted Word documents (.docx) and printable PDFs (.pdf) for offline classroom use.
Lucide React	`^0.562.0`	Comprehensive, lightweight vector icon set integrated across all dashboards and navigation sidebars.

4.2 Backend Architecture & Dependencies

Technology / Library	Version	Functional Role & Architectural Responsibility
Node.js & Express	`Node 22, Express 4.21`	High-throughput asynchronous REST API framework serving 16 modular route controllers and centralized error handling.
Prisma ORM	`5.22.0`	Type-safe database ORM managing 17 relational models, complex migrations, relation queries, and parameterized SQL execution.
@prisma/adapter-neon	`^5.22.0`	Specialized serverless PostgreSQL driver adapter for ultra-low latency connection pooling on Neon Serverless.
@google/generative-ai	`^0.24.1`	Official Google SDK orchestrating Gemini 2.5 Flash API calls, multimodal file attachments, and structured JSON generation.
Cloudflare R2 / AWS S3 SDK	`^3.968.0`	S3-compatible object storage SDK (`@aws-sdk/client-s3`) managing textbook uploads, exam PDFs, and CDN delivery.
JWT & bcryptjs	`jwt 9.0, bcrypt 2.4`	Cryptographic password hashing (12 salt rounds) and stateless JSON Web Token session validation with 7-day expiration.
Zod Validation	`^3.23.8`	Strict runtime schema validation across incoming request bodies, route parameters, and query strings.
Helmet & CORS	`helmet 8.0, cors 2.8`	Security hardening applying CSP headers, XSS prevention, HSTS policies, and explicit CORS origin white-listing.
Node-Cron	`^3.0.3`	Automated scheduled background tasks executing periodic database keepalive pings to eliminate cold starts.

5. The Savra AI Pedagogical Engine (Gemini 2.5 Flash)

The core intelligence powering Savra AI is encapsulated within `backend/src/lib/gemini.ts`, a dedicated pedagogical service exceeding 1,200 lines of rigorous prompt engineering, document injection logic, and structured JSON parsing. It harnesses Googleâ€™s **Gemini 2.5 Flash** model configured with low temperatures for deterministic educational outputs.

5.1 In-Context Pedagogical Grounding Architecture

Rather than relying on ungrounded model weights, Savra AI dynamically retrieves and injects official national framework documents directly into Gemini's multi-part context window as base64-encoded PDF payloads:

ðŸ“˜ Educational Framework Grounding

Loaded during lesson plan generation from `public/lesson-plan/`:

- **NEP_Final_English_0.pdf:** National Education Policy 2020 core tenets.
- **NCF-School-Education-Pre-Draft.pdf:** National Curriculum Framework 2023.
- **Blooms-Taxonomy.pdf:** Cognitive domains & action verbs.
- **The_5E_Model_of_Instruction.pdf:** Constructivist teaching stages.

ðŸ“‘ Official CBSE Question Paper Blueprints

Loaded during assessment generation from `public/question-paper/`:

- **Class 10 & 12 Board Papers:** Subject-specific official CBSE board papers (Maths, Science, Physics, Chemistry, Biology, English, etc.).
- **PARAKH & Evaluation Handbooks:** National assessment standardization guidelines.
- **CBSE 2025 QP Standardization:** Official section-by-section mark allocations.

5.2 Structured Lesson Plan Generation Engine

When a teacher requests a lesson plan for a grade, subject, and chapter, the backend invokes `generateLessonPlanPeriods()`. The generator produces a strictly structured JSON contract detailing every single teaching period:

Pedagogical Dimension	Content Generated per Classroom Period
Concept	Clear 2-3 sentence definition of the core scientific or humanities concept to be mastered.
Learning Outcomes	Specific, measurable student learning goals mapped to Bloom's taxonomy verbs (e.g., *Analyze*, *Differentiate*, *Formulate*).
Teacher-Learning Process	Cohesive 5E instructional sequence synthesizing Engagement, Exploration, Explanation, Elaboration, and Evaluation.
Formative Assessment	Rapid, informal comprehension check questions and peer-review tasks conducted within the period.
Teaching Resources	Required lab equipment, digital aids, blackboard diagrams, and reference literature.
21st-Century Skills & Values	Explicit mapping to critical thinking, collaboration, constitutional values, environmental ethics, and digital literacy.
Real-Life Applications	Tangible Indian and global real-world scenarios demonstrating practical relevance to everyday life.
Student Reflection	Metacognitive inquiry prompts encouraging students to self-assess their conceptual understanding.

5.3 CBSE-Compliant Question Paper & Assessment Engine

The assessment generator (`generateAssessment()`) produces formal examination papers adhering to CBSE sectional architecture:

- **Subject-Specific General Instructions:** Automatically embeds official board instructions (e.g., "Section A contains MCQs of 1 mark each", "Use of calculators is strictly prohibited").
- **9 Standardized Question Typologies:**
  1. *Multiple Choice Questions (MCQ):* 4 balanced options with single correct key.
  2. *Short Answer Questions:* 2-mark conceptual explanations (2-3 sentences).
  3. *Long Answer Questions:* 5-mark comprehensive analytical queries.
  4. *Case-Study / Source-Based Questions:* Contextual passages followed by sub-questions (a, b, c).
  5. *Fill in the Blanks:* Vocabulary and factual recall statements.
  6. *Diagram-Based Questions:* Labeling and drawing instructions for biological/physical systems.
  7. *Numerical Problems:* Multi-step mathematical calculations with formula credit.
  8. *Assertion-Reasoning (A-R):* Rigorous dual-statement logical evaluations.
  9. *Map-Based Questions:* Geographical and historical landmark labeling tasks.

5.4 Automated Concise Answer Key Generator

To eliminate teacher evaluation fatigue, `generateAnswerKey()` immediately parses the newly generated question paper and creates an ultra-concise, plain-text marking scheme. MCQs return the exact option letter and a 1-sentence rationale; numerical problems return final figures and formulas; long-form answers highlight 3 to 5 indispensable scoring criteria.

5.5 Ask Savra Multimodal AI Assistant

The interactive AI assistant (`generateAIResponse()` & `analyzeImage()`) dynamically swaps system prompts based on authenticated user roles:

- **Student Mode:** Operates as a friendly, patient Socratic tutor. It guides homework without providing copy-paste solutions, gives conceptual hints, and maintains child-safe boundaries.
- **Teacher Mode:** Operates as an experienced curriculum specialist. Suggests differentiated classroom strategies, remedial action plans, and analyzes uploaded textbook photos via Gemini Multimodal Vision.

6. Complete Database Schema & Data Models

The Savra AI persistence tier is managed via **Prisma ORM 5.22.0** connected to a **Neon Serverless PostgreSQL** cluster. The schema is normalized into 28 models and junction entities, enforcing referential integrity, cascading deletes, and optimized B-tree indexing.

6.1 Core Authentication & Profile Models

Model Name	Key Fields & Data Types	Relational Links	Operational Purpose & Constraints

`User`	`id (UUID PK)`
	`email (String Unique)`
	`passwordHash (String?)`
	`googleId (String?)`
	`role (Role Enum)`
	`name (String)`
	`avatarUrl (String?)`
		1:1 with `Teacher`, `Student`, `Admin`
		1:N with `File`, `AIConversation`
			Base identity record. Supports password authentication (bcrypt) and Google OAuth SSO. Supports soft deletes via `deletedAt`.

`School`	`id (UUID PK)`
	`name (String)`
	`code (String Unique)`
		1:N with `Class`, `Teacher`, `Admin`	Multi-tenant institutional anchor. Teachers and students bind to their school using the unique institutional code (e.g., `SAVRA001`).

`Teacher`	`id (UUID PK)`
	`userId (UUID FK Unique)`
	`schoolId (UUID FK)`
	`location (String?)`
	`teacherRole (String?)`
	`onboardingCompleted (Boolean)`
		Belongs to `User`, `School`
		M:N with `Subject`, `Class`
		1:N with `Lesson`, `Quiz`, `Assessment`, `Announcement`
			Represents educator profiles. Tracks onboarding status (subject and grade assignments) and cascades on user account deletion.

`Student`	`id (UUID PK)`
	`userId (UUID FK Unique)`
	`classId (UUID FK?)`
	`rollNumber (String?)`
	`totalPoints (Int default 0)`
		Belongs to `User`, `Class`
		M:N with `Subject`
		1:N with `QuizAttempt`, `AnnouncementRead`
			Represents learner profiles. Stores cumulative gamification coin points and links to their assigned classroom section.

`Admin`	`id (UUID PK)`
	`userId (UUID FK Unique)`
	`schoolId (UUID FK)`
		Belongs to `User`, `School`	Institutional oversight account with full administrative control over classrooms, teacher assignments, and rosters.

6.2 Academic Topology & Taxonomy Models

Model Name	Key Fields & Data Types	Relational Links	Operational Purpose & Constraints

`Class`	`id (UUID PK)`
	`schoolId (UUID FK)`
	`grade (Int 1-12)`
	`section (VarChar 1)`
	`name (String)`
		Belongs to `School`
		1:N with `Student`, `Lesson`, `Quiz`, `Assessment`, `Announcement`
			Composite unique constraint `[schoolId, grade, section]`. Represents physical classrooms (e.g., "Class 10 A").

`Subject`	`id (UUID PK)`
	`name (String)`
	`code (String Unique)`
		1:N with `Chapter`, `Lesson`, `Quiz`, `Assessment`
		M:N with `Teacher`, `Student`
			Represents standardized curriculum subjects (Maths, Science, English, Physics, Chemistry, Biology, History, Geography, etc.).

`Chapter`	`id (UUID PK)`
	`subjectId (UUID FK)`
	`name (String)`
	`grade (Int default 6)`
	`orderIndex (Int)`
		Belongs to `Subject`
		M:N with `Lesson`, `Quiz`, `Assessment`
			Composite unique constraint `[subjectId, name, grade]`. Organizes textbook syllabi sequentially for granular lesson mapping.

6.3 Lesson Planning & Content Persistence

Model Name	Key Fields & Data Types	Relational Links	Operational Purpose & Constraints

`Lesson`	`id (UUID PK)`
	`title (String)`
	`status (draft, saved, published)`
	`topic (Text?)`
	`numberOfPeriods (Int?)`
	`referenceFileUrl (String?)`
		Belongs to `Teacher`, `Class`, `Subject`
		1:N with `LessonPeriod`
		M:N with `Chapter`
			Represents master lesson plan units. Tracks planning progression from AI draft to published classroom curriculum.

`LessonPeriod`	`id (UUID PK)`
	`periodNo (Int)`
	`concept (Text?)`
	`learningOutcomes (Text?)`
	`teacherLearningProcess (Text?)`
	`assessment (Text?)`
	`resources (Text?)`
	`centurySkillsValueEducation (Text?)`
	`realLifeApplication (Text?)`
	`reflection (Text?)`
		Belongs to `Lesson` (Cascade Delete)	Composite unique constraint `[lessonId, periodNo]`. Stores the 8-dimensional pedagogical breakdown for every period.

6.4 Gamified Quizzes & Formal Examination Models

Model Name	Key Fields & Data Types	Relational Links	Operational Purpose & Constraints
Quiz	id (UUID PK) title (String) timeLimit (Int minutes) difficultyLevel (easy, medium, hard) totalQuestions (Int) totalMarks (Int) status (QuizStatus)	Belongs to Teacher, Class, Subject 1:N with Question, QuizAttempt M:N with Chapter	Interactive mobile/web quiz unit. Configured with optional due dates and reference file attachments for generation grounding.
Question & QuestionOption	questionText (Text) questionType (String) marks (Int default 1) optionLabel (VarChar 1) optionText (Text) isCorrect (Boolean)	Question belongs to Quiz Options belong to Question Answers belong to Question	Stores quiz questions and four corresponding labeled options (A, B, C, D) with deterministic correctness flags.
QuizAttempt & StudentAnswer	score (Int?) percentage (Decimal?) timeTaken (Int seconds) status (in_progress, submitted, graded)	Attempt belongs to Quiz, Student StudentAnswer belongs to Attempt, Question	Records student submissions. Tracks time elapsed in seconds, auto-scores MCQs, and triggers gamified point increments.
Assessment & QuestionType	title (String) totalMarks (Int) questionPaper (JSON) questionType (String) numberOfQuestions (Int)	Belongs to Teacher, Class, Subject 1:N with AssessmentQuestionType	Represents formal CBSE exam papers. Stores the complete structured examination paper JSON including sections and instructions.
Announcement & Read	title (String) content (Text) attachmentUrl (String?) readAt (DateTime)	Announcement belongs to Teacher, Class Read belongs to Announcement, Student	Classroom bulletin board. Tracks which individual students have read each notice via unique compound index [announcementId, studentId].
AIConversation & AIMessage	title (String?) role (user | assistant) content (Text) fileUrl (String?)	Conversation belongs to User Message belongs to Conversation	Stores Ask Savra chat sessions. Retains historical context for multi-turn inquiry and multimodal attachment analysis.

7. User Flows & Operational Journeys

7.1 Teacher Operational Workflow

Step 1: Auth & Guided Onboarding

Teacher signs up using email/password or Google OAuth. Selects institutional school code (e.g., `SAVRA001`). Selects assigned classes (e.g., Class 10 A) and primary subject faculties. Onboarding status transitions to `true`.

Step 2: AI-Powered Lesson Planning

Teacher specifies grade, subject, and syllabus chapters. Selects target number of classroom periods and pedagogical objectives. Clicks "Generate with Savra AI". Gemini injects 5E, NEP, NCF, and Bloom's frameworks to create the period table in seconds.

Step 3: In-Browser Customization & Export

Teacher reviews generated periods in the interactive TipTap editor. Customizes learning outcomes, adjusts instructional resources, and saves changes. Exports document as formatted Microsoft Word (`.docx`) or printable PDF for offline submission.

Step 4: Examination Paper Generation

Teacher navigates to Assessment Creator. Specifies question distribution (e.g., 10 MCQs, 4 Short Answers, 2 Case Studies). Savra AI cross-references official Class 10/12 CBSE sample papers to generate the question paper and produces a concise answer key.

Step 5: Gamified Quiz Publishing

Teacher initiates Quiz Creator. Selects difficulty (Easy, Medium, Hard) and time limit (e.g., 15 minutes). Gemini generates MCQs with smart distractors. Teacher reviews questions and publishes directly to the designated class section.

Step 6: Class Analytics & Remedial Intervention

Teacher monitors the real-time Analytics Dashboard. Reviews class average scores, individual student completion rates, and the "Top Missed Questions" chart. Uses AI insights to schedule targeted remedial revision periods.

7.2 Student Operational Workflow

Step 1: Enrollment & Dashboard Entry

Student registers with school code and classroom section. Lands on the personalized Student Home showing total coin balance, class rank, pending quizzes, and unread teacher announcements.

Step 2: Gamified Quiz Participation

Student selects an available quiz. Starts interactive timed attempt. Answers MCQs with progress indicators and instant visual feedback. Submits attempt before timer expiration.

Step 3: Instant Score & Points Reward

Backend instantly scores the attempt. Student sees percentage score, correct answers, and explanation tooltips. Earned points are automatically credited to `Student.totalPoints`.

Step 4: Leaderboard & Socratic Tutoring

Student views class-wide and school-wide Leaderboards. If struggling with a concept, opens "Ask Savra" AI Tutor for step-by-step conceptual hints without direct answer revelation.

7.3 School Administrator Operational Workflow

Institutional Setup & Topology

Admin manages the institutional master record. Verifies classroom sections across Grades 1 to 12. Oversees school code generation and academic year parameters.

Faculty & Roster Supervision

Admin audits the active teacher roster. Assigns subjects and classroom responsibilities. Manages student enrollments, transfers between sections, and roll number allocations.

Academic Output Auditing

Admin accesses institutional analytics: total lesson plans generated, examination papers published, quiz engagement across grades, and faculty adoption metrics.

System Health & Compliance

Ensures curriculum compliance across all school sections in accordance with CBSE circulars and state board regulations.

8. Exhaustive REST API Catalog

The Savra AI backend provides a uniform, versioned RESTful API under the `/api` prefix. All routes require `Bearer <JWT>` authentication unless marked public. Requests are validated via Zod schemas, and responses follow the standard envelope: `{ success: boolean, data?: any, error?: { message: string, code: string } }`.

8.1 Authentication & User Identity Routes (`/api/auth`)

Method	Endpoint Path	Access Role	Functionality & Validation
POST	`/api/auth/register`	Public	Registers a new user (Teacher, Student, or Admin). Hashes password with bcrypt (12 rounds) and creates linked role profile.
POST	`/api/auth/login`	Public	Authenticates email/password. Issues stateless JWT access token (7-day validity) and returns user profile payload.
POST	`/api/auth/google`	Public	Verifies Google OAuth ID token. Creates user account if first-time login, or issues session JWT for existing accounts.
GET	`/api/auth/me`	Authenticated	Returns currently authenticated user profile, active role, linked school code, and profile metadata.
POST	`/api/auth/logout`	Authenticated	Terminates user session client-side and logs session completion.
POST	`/api/auth/forgot-password`	Public	Generates cryptographically secure password reset token and dispatches reset instructions.
POST	`/api/auth/reset-password`	Public	Validates reset token and sets new bcrypt password hash for user account.

8.2 Lesson Planning Routes (`/api/lessons`)

Method	Endpoint Path	Access Role	Functionality & Validation
POST	`/api/lessons`	Teacher	Creates a new lesson record associated with teacher, class, subject, and multiple chapters.
GET	`/api/lessons`	Teacher	Lists lessons created by the teacher with filtering by classId, subjectId, and status (draft, saved, published).
GET	`/api/lessons/:id`	Teacher	Fetches single lesson record including all associated periods (1 to N) and linked chapter metadata.
PUT	`/api/lessons/:id`	Teacher	Updates lesson header, objectives, duration, and saves edited period content from the TipTap editor.
DELETE	`/api/lessons/:id`	Teacher	Deletes lesson record and cascades delete to all child `LessonPeriod` and junction entries.
POST	`/api/lessons/generate`	Teacher	Triggers Gemini AI engine with 5E, NEP, NCF, and Bloom's PDF grounding to generate multi-period structured lesson plan.
PATCH	`/api/lessons/:id/status`	Teacher	Updates lifecycle state of a lesson (`draft` → `saved` → `published`).
POST	`/api/lessons/generate-topic`	Teacher	AI generates concise 2-4 word pedagogical topic name based on selected chapters and grade level.

8.3 Formal Assessment Routes (`/api/assessments`)

Method	Endpoint Path	Access Role	Functionality & Validation
POST	`/api/assessments`	Teacher	Creates assessment record with question types distribution and target marks.
GET	`/api/assessments`	Teacher	Lists teacher's created question papers with pagination and subject/class filters.
GET	`/api/assessments/:id`	Teacher	Fetches complete assessment details, question distribution breakdown, and structured question paper JSON.
PUT	`/api/assessments/:id`	Teacher	Updates question paper content, modifies individual questions, marks, and section titles.
DELETE	`/api/assessments/:id`	Teacher	Deletes assessment record and all associated question type allocations.
POST	`/api/assessments/generate`	Teacher	Generates CBSE-aligned question paper by injecting official Class 10/12 sample papers and guidelines into Gemini.
POST	`/api/assessments/generate-answer-key`	Teacher	Parses generated question paper and creates an ultra-concise, plain-text marking scheme and answer key.
PATCH	`/api/assessments/:id/status`	Teacher	Transitions assessment status (`draft` → `saved` → `published`).

8.4 Gamified Quiz Routes (`/api/quizzes`)

Method	Endpoint Path	Access Role	Functionality & Validation
POST	`/api/quizzes`	Teacher	Creates a new gamified quiz container with time limits, due dates, and difficulty settings.
GET	`/api/quizzes`	Teacher	Lists quizzes created by the teacher with status filters and question counts.
GET	`/api/quizzes/:id`	Teacher / Student	Retrieves quiz details including all questions and option choices (masks correct answers for students).
POST	`/api/quizzes/generate`	Teacher	AI generates multiple-choice questions with 4 options, balanced distractors, and difficulty calibration.
GET	`/api/quizzes/:id/attempts`	Teacher	Lists all student attempts for a quiz, showing individual scores, percentages, and time taken.
POST	`/api/quizzes/:id/questions`	Teacher	Appends custom manual questions and option sets to an existing quiz.
PUT	`/api/quizzes/:id/questions/:qId`	Teacher	Updates question text, marks, and option labels/correctness flags.

8.5 Student Dashboard Routes (`/api/student`)

Method	Endpoint Path	Access Role	Functionality & Validation
GET	`/api/student/dashboard`	Student	Returns student overview: total coin points, class rank, total attempts, pending quizzes, and unread notices.
GET	`/api/student/quizzes/available`	Student	Lists all published quizzes assigned to the student's class that have not yet been completed.
POST	`/api/student/quizzes/:id/start`	Student	Initializes a new `QuizAttempt` in `in_progress` state. Starts server-side attempt timestamp.
POST	`/api/student/quizzes/:id/attempt/:aId/answer`	Student	Saves individual question answer choice during an in-progress attempt.
PATCH	`/api/student/quizzes/:id/attempt/:aId/submit`	Student	Submits quiz attempt. Auto-grades MCQs, calculates percentage score, and awards earned points to `Student.totalPoints`.
GET	`/api/student/leaderboard`	Student	Returns real-time class-wide and school-wide leaderboard standings ranked by total points.
GET	`/api/student/announcements`	Student	Fetches class announcements with read status indicators and downloadable teacher attachments.
POST	`/api/student/announcements/:id/read`	Student	Upserts `AnnouncementRead` record timestamping when the student viewed the notice.

8.6 Teacher & Admin Hub Routes (`/api/teacher` & `/api/admin`)

Method	Endpoint Path	Access Role	Functionality & Validation
GET	`/api/teacher/profile`	Teacher	Returns teacher profile, assigned subjects, and active classroom sections.
POST	`/api/teacher/onboarding`	Teacher	Completes teacher onboarding transaction: binds school, assigned subjects, and teaching classes.
GET	`/api/teacher/analytics/overview`	Teacher	Aggregates teacher classroom metrics: total quizzes taken, pass rates, average marks, and engagement trends.
GET	`/api/admin/dashboard`	Admin	Provides school-wide executive analytics: total teachers, enrolled students, active classrooms, and content creation volume.
GET	`/api/admin/teachers`	Admin	Lists all faculty members in the school with their assigned classes, subjects, and activity status.
POST	`/api/admin/teachers`	Admin	Creates a new teacher account and allocates initial subject and classroom delegations.
GET	`/api/admin/students`	Admin	Lists all enrolled students with roll number filtering, class breakdown, and performance metrics.
POST	`/api/admin/students/transfer`	Admin	Transfers a student between classroom sections or updates their assigned roll number.

8.7 Multimodal AI & Cloud Storage Routes (`/api/ai` & `/api/upload`)

Method	Endpoint Path	Access Role	Functionality & Validation
POST	`/api/ai/query`	Teacher / Student	Sends a chat prompt to Ask Savra AI with rate-limiting (20 req/min) and historical context window management.
POST	`/api/ai/analyze-image`	Teacher / Student	Multimodal vision endpoint: analyzes uploaded math problem images or textbook diagrams using Gemini 2.5 Flash.
GET	`/api/ai/conversations`	Authenticated	Lists past chat sessions with auto-generated conversation titles.
POST	`/api/upload/file`	Authenticated	Uploads files to Cloudflare R2 / AWS S3 using multipart form data (Multer memory storage). Returns CDN URL.
GET	`/api/upload/file/*`	Public	Secure proxy streaming endpoint serving R2 assets with 1-year immutable caching headers (`Cache-Control: public, max-age=31536000`).
GET	`/api/health`	Public	Health check endpoint returning system status and ISO timestamp for monitoring uptime.

9. Security, Data Privacy & Reliability Engineering

9.1 Authentication & Role-Based Access Control (RBAC)

Security is enforced at every layer of the Savra AI infrastructure:

- **Stateless Cryptographic Tokens:** User sessions are managed via digitally signed JWTs containing `userId`, `email`, and `role` (`teacher`, `student`, `admin`). Tokens expire after 7 days, eliminating database session lookup bottlenecks.
- **Granular Authorization Middleware:** The `authorize(...roles)` guard inspects token claims before route handler execution, rejecting unauthorized access with `403 Forbidden` responses.
- **Password Hashing:** User passwords are encrypted using `bcryptjs` configured with **12 salt rounds**, providing robust defense against rainbow tables and brute-force cracking.

9.2 Strict Data Validation & SQL Injection Immunity

ðŸ›¡ï¸ Zod Schema Validation

All incoming payloads (bodies, queries, params) must pass strict Zod schemas before reaching business logic. Malformed requests are rejected immediately with descriptive `400 Bad Request` error structures.

ðŸ”’ SQL Injection Immunity

Prisma ORM compiles all database queries into parameterized SQL statements. User inputs are never concatenated directly into raw queries, rendering SQL injection vectors impossible.

â±ï¸ AI Abuse Mitigation

In-memory rate limiting throttles AI query endpoints to a maximum of **20 requests per minute per user**, safeguarding API quotas and preventing denial-of-wallet attacks.

ðŸ”„ Database Keepalive Cron

A background cron job (`node-cron`) executes a lightweight `SELECT 1` ping every 5 minutes, preventing serverless cold starts on Neon PostgreSQL and Render backend instances.

10. Deployment, Environment Variables & Configuration

10.1 Environment Variables Reference

Variable Name	Default / Sample	Description & Functional Purpose
DATABASE_URL	postgresql://...	Neon Serverless PostgreSQL connection string with SSL mode enabled.
JWT_SECRET	[cryptographic string]	Secret key used for signing and verifying JSON Web Tokens.
GEMINI_API_KEY	AIzaSy...	Google AI Studio API key enabling Gemini 2.5 Flash generative models.
AI_MODEL	gemini-2.5-flash	Specifies the generative model identifier for lesson planning and assessment generation.
AI_MAX_TOKENS	1500	Maximum token output limit for conversational responses in Ask Savra chat.
AI_RATE_LIMIT_PER_MINUTE	20	Rate limit threshold controlling maximum AI queries allowed per minute per user.
R2_ACCOUNT_ID	[cloudflare account]	Cloudflare account ID for S3-compatible R2 object storage access.
R2_ACCESS_KEY_ID	[access key]	Cloudflare R2 authentication key ID for bucket read/write operations.
R2_SECRET_ACCESS_KEY	[secret key]	Cloudflare R2 authentication secret key for secure signing.
R2_BUCKET_NAME	savra-ai	Target storage bucket name housing uploaded textbook chapters and reference materials.
KEEPALIVE_INTERVAL_MINUTES	5	Interval in minutes between automated database ping operations to eliminate cold starts.
FRONTEND_URL	http://localhost:3000	Whitelisted CORS origin allowing browser requests from the Next.js frontend client.

11. Future Roadmap & Strategic Vision

Savra AI is positioned to become the premier operating system for Indian school education. Key initiatives planned for subsequent development cycles include:

ðŸŒ Multilingual Regional Localization

Expanding beyond English and Hindi to support 10+ major regional Indian languages (Tamil, Telugu, Kannada, Malayalam, Bengali, Marathi, Gujarati) for vernacular state board schools.

âœï¸ Vision-Based Handwritten Answer Grading

Leveraging Gemini Multimodal Vision to automatically scan, OCR, and evaluate handwritten student examination sheets, providing automated point grading and margin annotations.

ðŸ“² WhatsApp Parent Progress Integration

Automated bi-weekly WhatsApp progress digests sent to parents, highlighting student quiz mastery, completed competencies, and recommended home revision topics.

ðŸŽ™ï¸ Voice-Activated Classroom Co-Pilot

Microphone integration for classroom smart interactive displays, allowing teachers to summon diagrams, real-time analogies, and comprehension check quizzes via hands-free voice commands.

Savra AI Educational Technologies â€¢ Built with â¤ï¸ for Indian Teachers & Students â€¢ CBSE & NEP 2020 Aligned